

Chapter 4

Production and Labor

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The Real Economy

- › We now begin our dive into macroeconomic modeling
 - Economists use complex models as it's generally hard (and usually unethical) to run nation-wide experiments
- › The first model we'll build is a *Real* Business Cycle model
 - But first, we need to define its parts

Partial Equilibrium

- › **Partial equilibrium (PE)** *not* all markets are in equilibrium
- › **General equilibrium (GE)** all markets are in equilibrium
- › We'll study PE in individual markets first, then move to GE
 1. Labor market
 2. Goods/loanable funds market(s)
 3. Money/asset market(s) – post exam 1

Partial Equilibrium in the Labor Market

1. Discuss basics of production
 2. Microfoundations of labor demand & profit maximization
 3. Microfoundations of labor supply & utility maximization
 4. Market clearing (supply + demand)
 5. Aggregate summary
- › Sets up how we'll study goods & money markets later

Factors of Production

- › **Factors of production** are resources used to make goods and services

- › Inputs to production that make outputs:
 - Labor (N)
 - Capital (K)
 - Other: land, financing, energy, entrepreneurship

Relationship b/t Inputs and Output

- › One way to describe the relationship between inputs and output is with a function:

$$Y = zF(K, N)$$

- › Where: Y = output

z = “total factor productivity, TFP” (technology)

K = capital

N = labor

Functional Form

- › Usually, we assume $F()$ is a Cobb–Douglas function

$$Y = zK^{\alpha}N^{\nu}$$

- › Where: α = capital share of output
 ν = labor share of output
- › Easy to find total Y , K , N and α , ν from data
 - We can then back out TFP

Cobb–Douglas and Returns to Scale

- › **Returns to scale:** how much more output do we get when we increase an input?
- › Increasing returns to scale:
 - Double *all* inputs, get more than double the output
- › Constant returns to scale:
 - Double *all* inputs, get exactly double the output
- › Decreasing returns to scale:
 - Double *all* inputs, get less than double the output

Identifying Returns to Scale

- › Look at sum of exponents:
 - $\alpha + \nu > 1 \rightarrow$ increasing returns
 - $\alpha + \nu = 1 \rightarrow$ constant returns
 - $\alpha + \nu < 1 \rightarrow$ decreasing returns

- › Typical U.S. firm α and ν estimates are 0.27 and 0.62
 - Implying DRS at the firm-level
 - Aggregate economy is (roughly) CRS

The Representative Agent

- › For now: only 1 firm that represents all firms in our model
 - This is the representative firm (or household)
 - Further assume 1 good produced (thus only 1 price)

- › Usually for quantities, capital letters denote aggregates; lower case letters denote firm/household level variables
 - But since there is only 1 representative firm, $K = k$
 - › I would love to say this notation is always true, but it is the economist's job to make the obvious opaque, so this may change

Profit Maximization: Labor and Capital

- › Firms pick labor and capital for profit maximization
 - Ch. 2 assumption: firms only care about maximizing value
 - Remember: firms demand labor; HHs supply labor

Optimal quantity: marginal benefit = marginal cost

Marginal Productivity

- › To figure out the marginal benefit of inputs, we need their marginal productivities
 - Productivity of k and n are the benefits to firm value
 - This chapter will focus on labor, but the general approach applies for capital
- › **Marginal Productivity of Labor (MPN):**
- › Change in output from adding 1 more unit of labor
- › **Marginal Productivity of Capital (MPK):**
- › Change in output from adding 1 more unit of capital

Marginal Productivity of Labor

› Change in output from adding 1 more unit of labor:

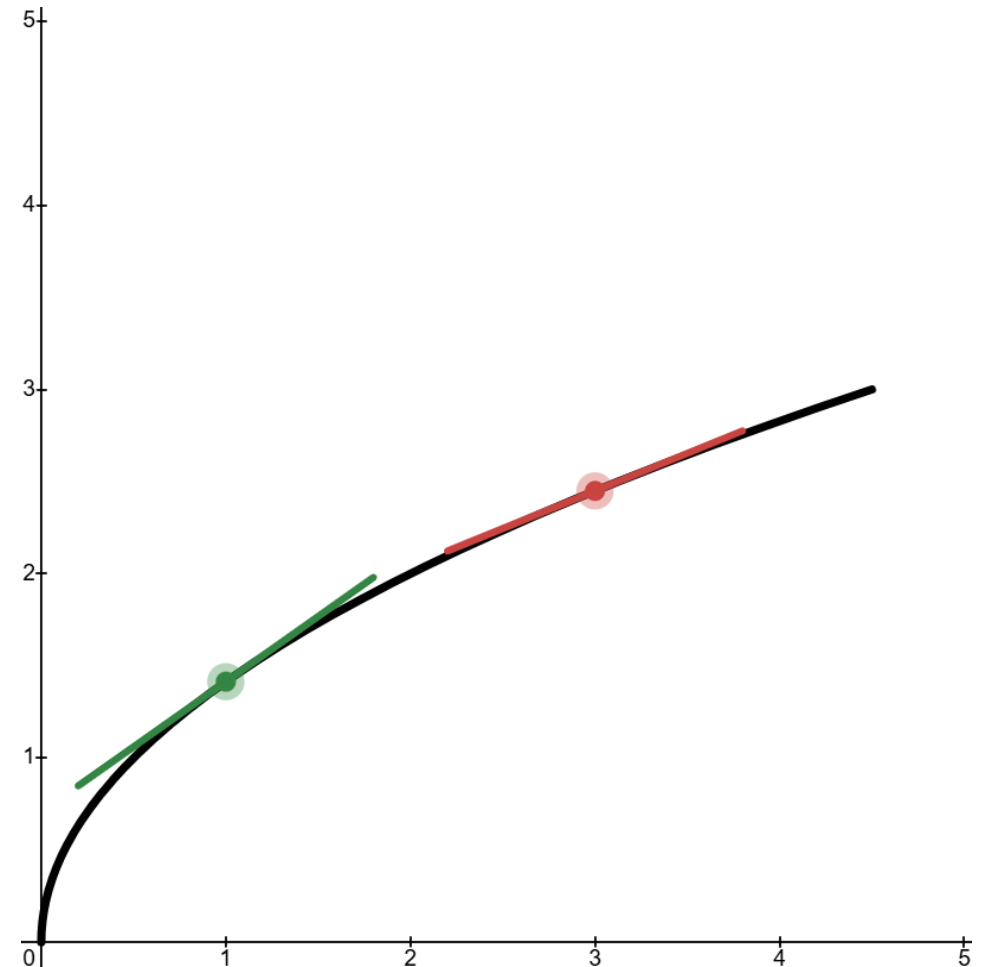
$$\frac{\Delta Y}{\Delta N}, \text{ where } \Delta \text{ denotes change}$$

› Alternatively, one can use calculus to find:

$$\frac{dY}{dN}$$

Slope

- › These equations give us the slope at a point
 - Rate of change at a point
 - How Y changes with N
- › Notice the slope gets flatter as the x-value rises

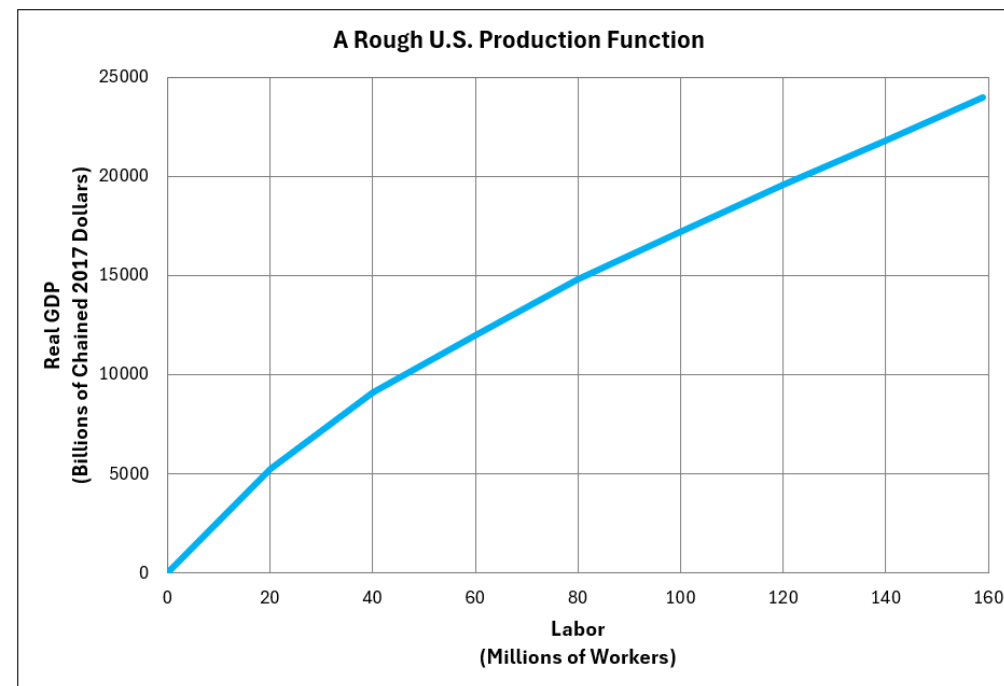


Properties of Marginal Product of Labor

1. MPN is positive: Y increases with more labor
2. But MPN decreases as N increases
 - › **Law of diminishing returns:**
 - *Additional* output falls as inputs increase
 - You get a lot of homework done with 1 computer
 - Not much additional homework done with 10 computers
 - “Too many chefs spoil the stew”
 - › Applies to K , as well!

Diminishing Production on a Graph

- › As N rises, Y rises
 - More capital = more output
- › But it starts to flatten out in the limit
 - Can't grow forever
 - › Yes, technically, you still grow some epsilon (arbitrarily small) amount



Data: All Employees, Total Nonfarm (BLS, 2026) and Real Gross Domestic Product (BEA, 2026)

The Wage

- › The last ingredient is the marginal cost
- › The [nominal] **Wage (W)** is the price of a unit of labor
 - For simplicity we'll assume no worker benefits
- › The **Real Wage (w)** is the nominal wage in terms of the output good ($\frac{W}{P}$)
 - Purchasing power of wages
- › *Note: for prices, capital letters denote nominal variables; lower case letters denote real variables*

Profit Maximization thru Labor

- › Firm's labor demand for profit maximization:

$$MPN = w$$

- › This ensures profit maximization for the firm
 - Also called the Firm “optimality” or “market clearing” condition

Profit Maximization Example (1/3)

- › Assume the economy produces a consumption good, c , and the real wage is 2.00

Y	N	MPN	w	Profit
0	0	-	2	0
7	1		2	
11	2		2	
13	3		2	
14	4		2	
14	5		2	

Profit Maximization (2/3) – Filling out *MPN*

Y	N	MPN	w	Profit
0	0	-	2	0
7	1	7	2	
11	2	4	2	
13	3	2	2	
14	4	1	2	
14	5	0	2	

Profit Maximization (3/3) – Finding Profit

Y	N	MPN	w	Profit
0	0	-	2	0
7	1	7	2	5
11	2	4	2	7
13	3	2	2	7
14	4	1	2	6
14	5	0	2	4

$$\text{Real Profits} = Y - w \times N$$

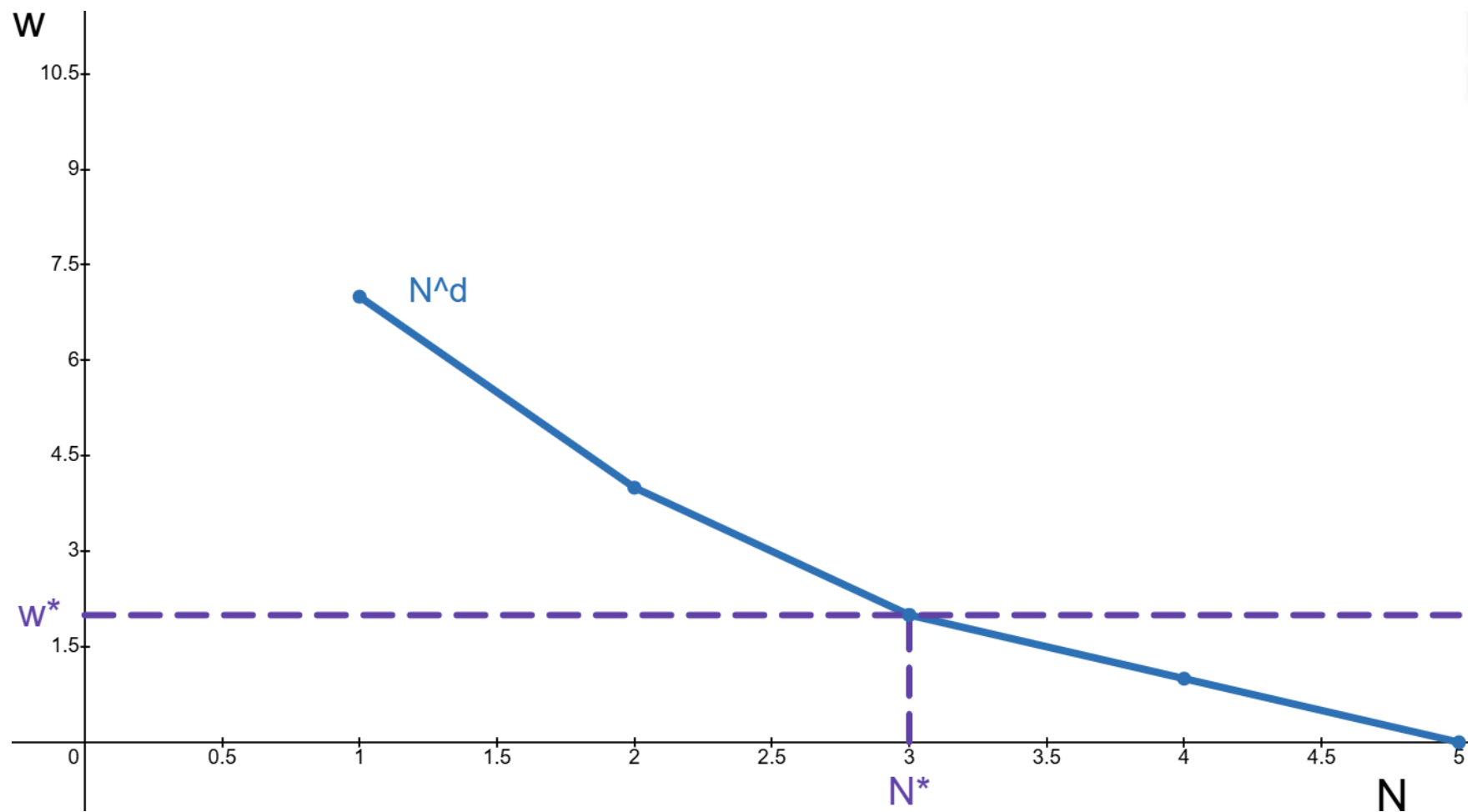
- The same analysis works for the nominal wage, too (multiply by P).

A Note on 1-good Economies

- › We study a 1 good economy for simplicity
 - Or a variable that represents the average basket of goods
- › 1 output good $\rightarrow$ 1 output price
 - Don't need CPI, PCE to determine price level
 - The price level is the 1 good's price

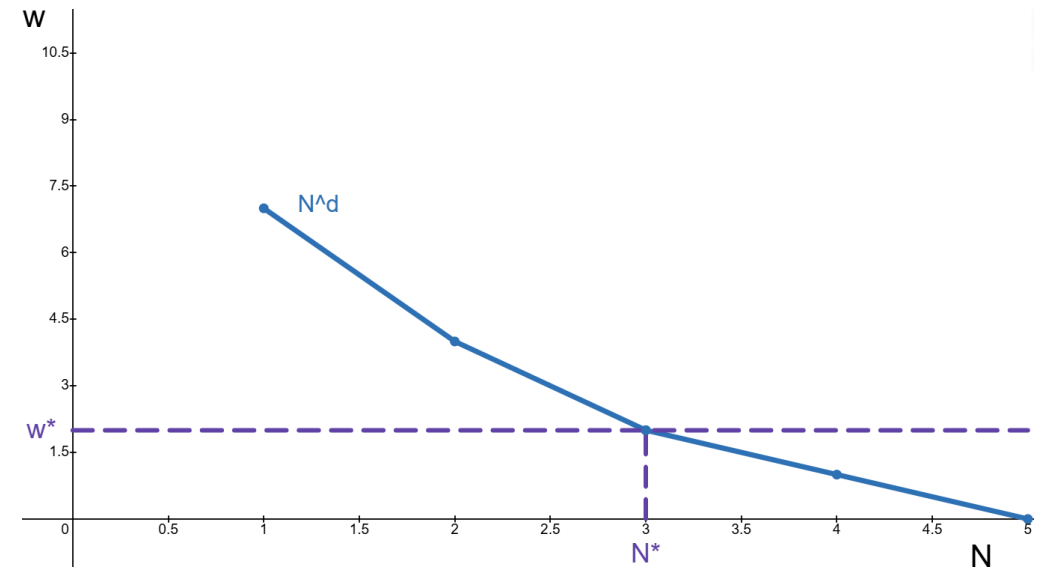
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Graphing Total Labor Demand



Interpreting the Graph

- › The points on the curve are the *MPN* values
 - $N^d = NPN$
- › If all labor becomes more productive, *MPN* $\uparrow$
 - $N^d \uparrow$
- › If the firm wants to lower *MPN*, it increases N



The Supply of Labor

- › The supply of labor comes from households
- › Two starting assumptions for households:
 - HHs like **leisure** (ℓ), time not spent working
 - › Equivalent to saying they dislike labor
 - They like consumption (c), but need labor income to afford it

Time Endowment Constraint

- › The HH is endowed with some amount of time each day
 - For simplicity, normalize to 1
 - They have 100% of their day to divide into labor or leisure
- › Describe ℓ and n^s in terms of *fraction of hours* in the day:

$$1 = \ell + n^s$$

- Easier to substitute away a variable to shrink problem
- HH gets utility from leisure, not labor

Extensive and Intensive Labor Choice

- › Two choices for labor: extensive and intensive
- › **Extensive labor choice:** Do you work? Yes or no
 - Only 1 of 2 answers
- › **Intensive labor choice:** How much do you work?
 - Continuum of answers
- › We'll assume everyone is making an intensive labor choice

The Household's Problem

› In real terms, the HH's problem is:

$$\max\{U(c, \ell)\} \text{ st: } c < w \times (1 - \ell) + X$$

- Maximize utility from c and ℓ , *subject to* their budget
- Budget is (wage $\times$ hours), plus income from other sources, X
 - X = NFP, taxes, transfers, asset returns, etc
- HH cannot spend more than their total income

Analyzing the Household's Budget

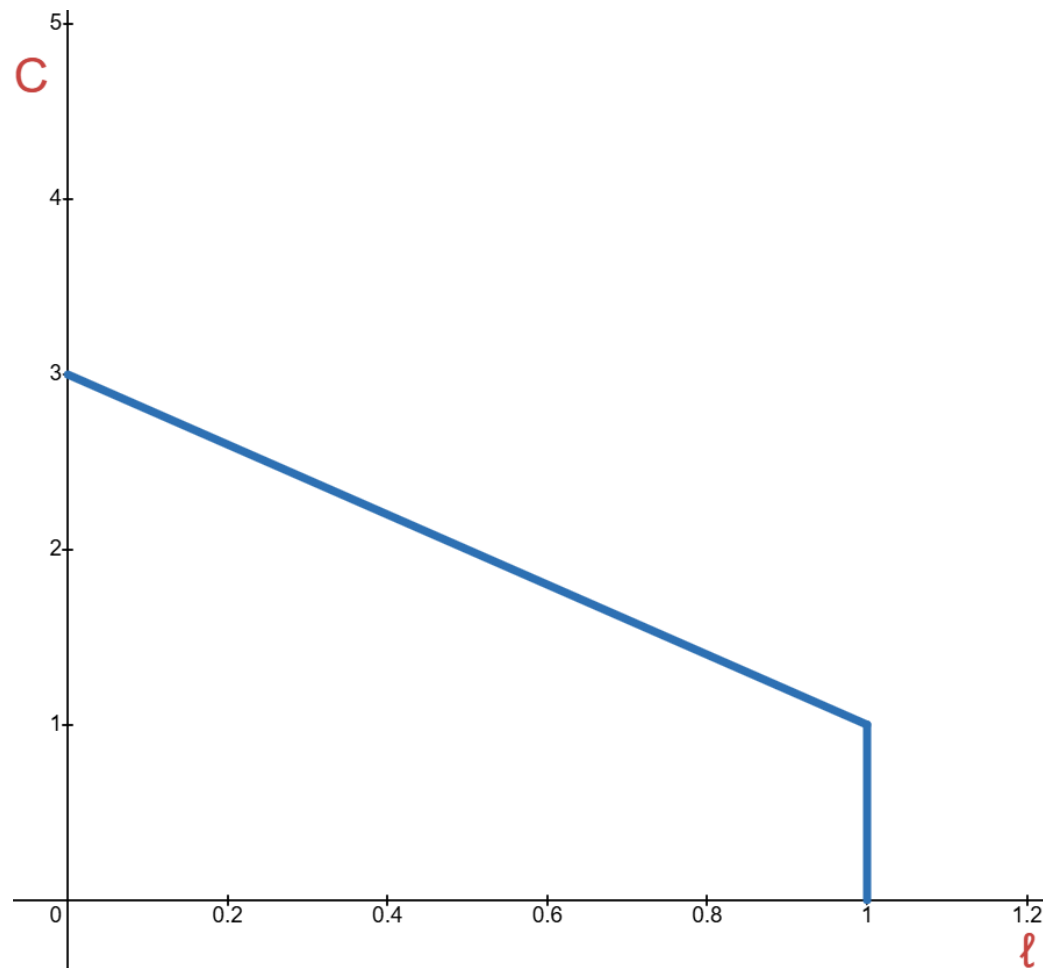
- › The budget is: $c < w \times (1 - \ell) + X$
- › We can assume equality: $c = w \times (1 - \ell) + X$
 - Why?
- › What does this equation look like now?

Relation to Slope-intercept Equation

- › $c = w \times (1 - \ell) + X$
 - c : y-axis
 - ℓ : x-axis
 - $-w$: slope
- › X is tricky to pinpoint; not the y-intercept
- › X = amount of c HH can afford without working
 - Consider when $\ell = 1$

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Graphing HH Budget



Labor-Leisure Trade-off

- › What does the HH choose?
 - Again, want marginal benefit = marginal cost

- › Marginal benefit of leisure:
 - Marginal utility of leisure: $MU(\ell)$

- › Marginal cost of leisure, in utility terms:
 - Give up some wages and potential c
 - Marginal utility of *affordable* c given up: $w \times MU(c)$

Utility Maximization thru Labor

- › HH chooses leisure/supplies labor for utility maximization:

$$MU(\ell) = w \times MU(c)$$

- › Commonly written as: $\frac{MU(\ell)}{MU(c)} = w$
 - Also called the HH “optimality” or “market clearing” condition
- › Like MPN and MPK , we assume MU diminishes

Marginal Rate of Substitution

- › **Marginal Rate of Substitution ($MRS_{x,y}$)** defines the rate at how a consumer will trade good x for good y
 - We know the two “goods” are ℓ and c
- › From the last equation:

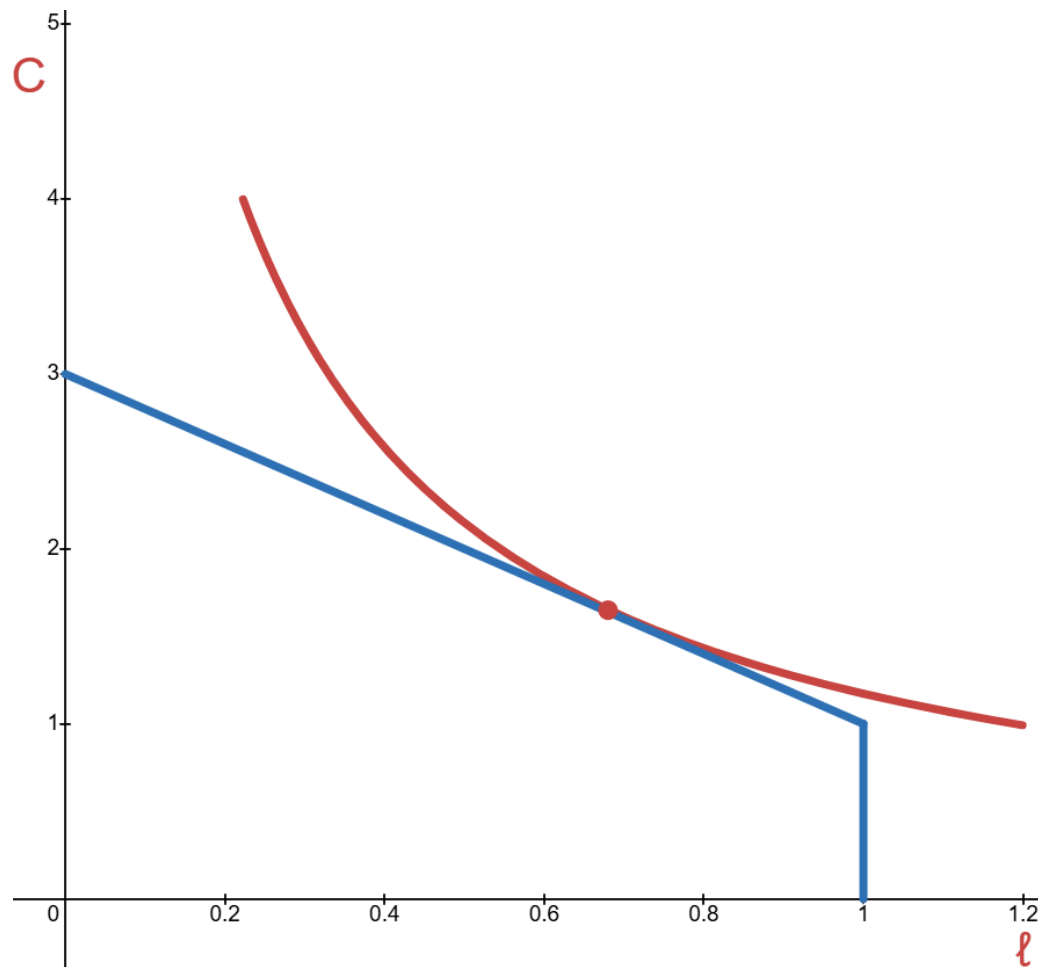
$$MRS_{\ell,c} = \frac{MU(\ell)}{MU(c)} = w$$

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Proof (Not required)

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Graphing HH's Choice



Relative Price

- › The real wage is the **relative price** of leisure
 - Price of one good in terms of another
 - By consuming more leisure, you give up the consumption you could have had with that labor income

$$\frac{MU(\ell)}{MU(c)} = w$$

- Real wage must afford enough consumption to make up for the utility lost by giving up leisure
- HH is indifferent between extra unit of c or ℓ

Real Wage and Intensive Labor Choice

- › An *increase* in the real wage can affect this in two ways:
 - Income effect*
 - Substitution effect

- › **Income effect:** you are richer; you consume more goods
 - Leisure is treated as a good; *you work less*

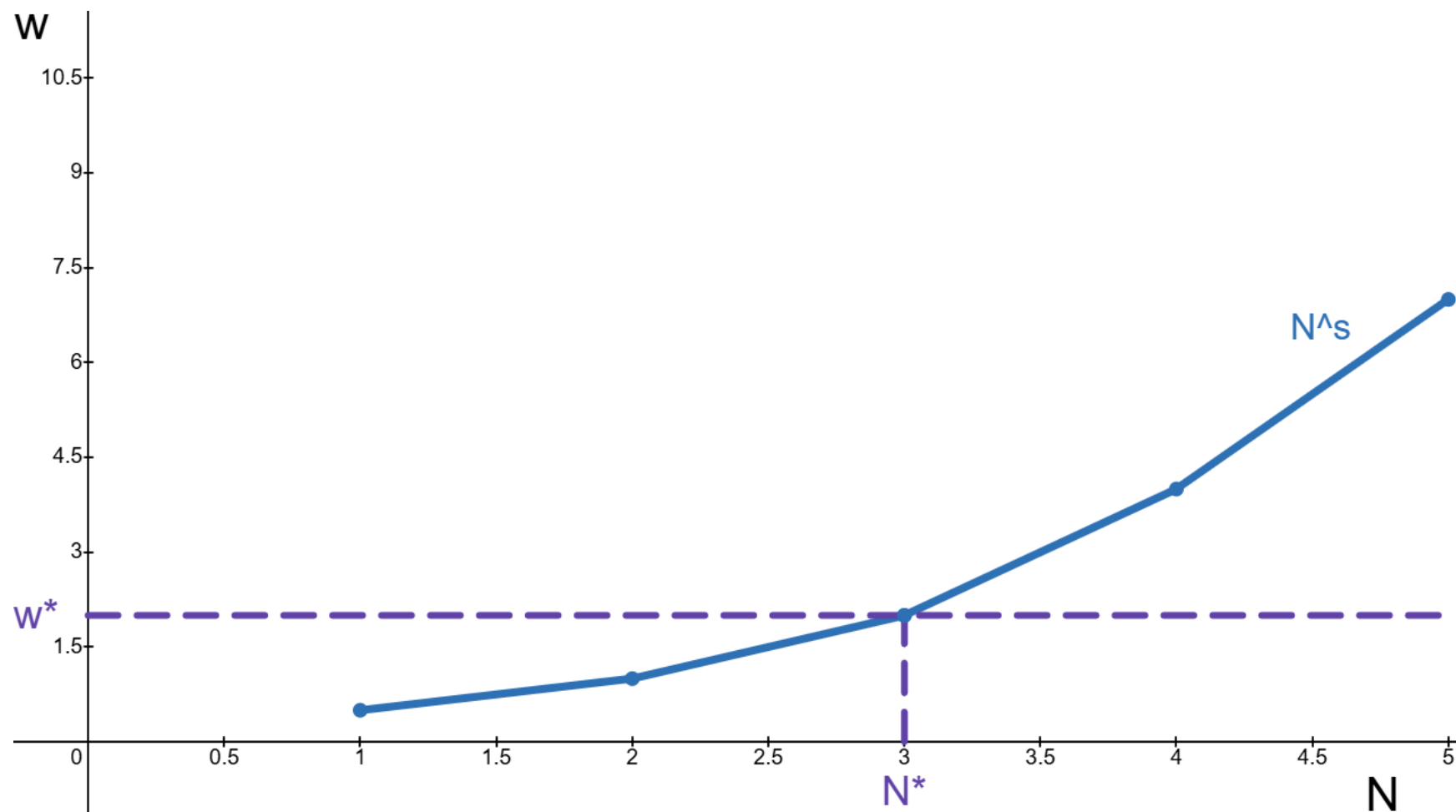
- › **Substitution effect:** relative price of leisure is higher
 - Substitute away from expensive good; *you work more*

- › *Recall through our lens, income increases wealth. However, labeling this as a wealth effect is awkward.

Substitution > Income

- › In general, we assume the substitution effect dominates
 - Increase in the real wage makes people work more
 - Labor supply curve slopes up
 - › At very high wage or wealth levels, the income effect may dominate. Why?

Graphing Real Labor Supply



Marginal Rate of Transformation

- › Before we move to equilibrium, we have to define one more rate
- › **Marginal Rate of Transformation ($MRT_{x,y}$)** defines the rate at how one good is transformed into another
 - We know the two “goods” are ℓ and c
- › From the firm’s condition:

$$MRT_{\ell,c} = MPN = w$$

HH and Firm Condition(s)

$$\frac{MU(\ell)}{MU(c)} = MPN = w$$

$$MRS_{\ell,c} = MRT_{\ell,c} = w$$

- › This condition ensures labor market clearing

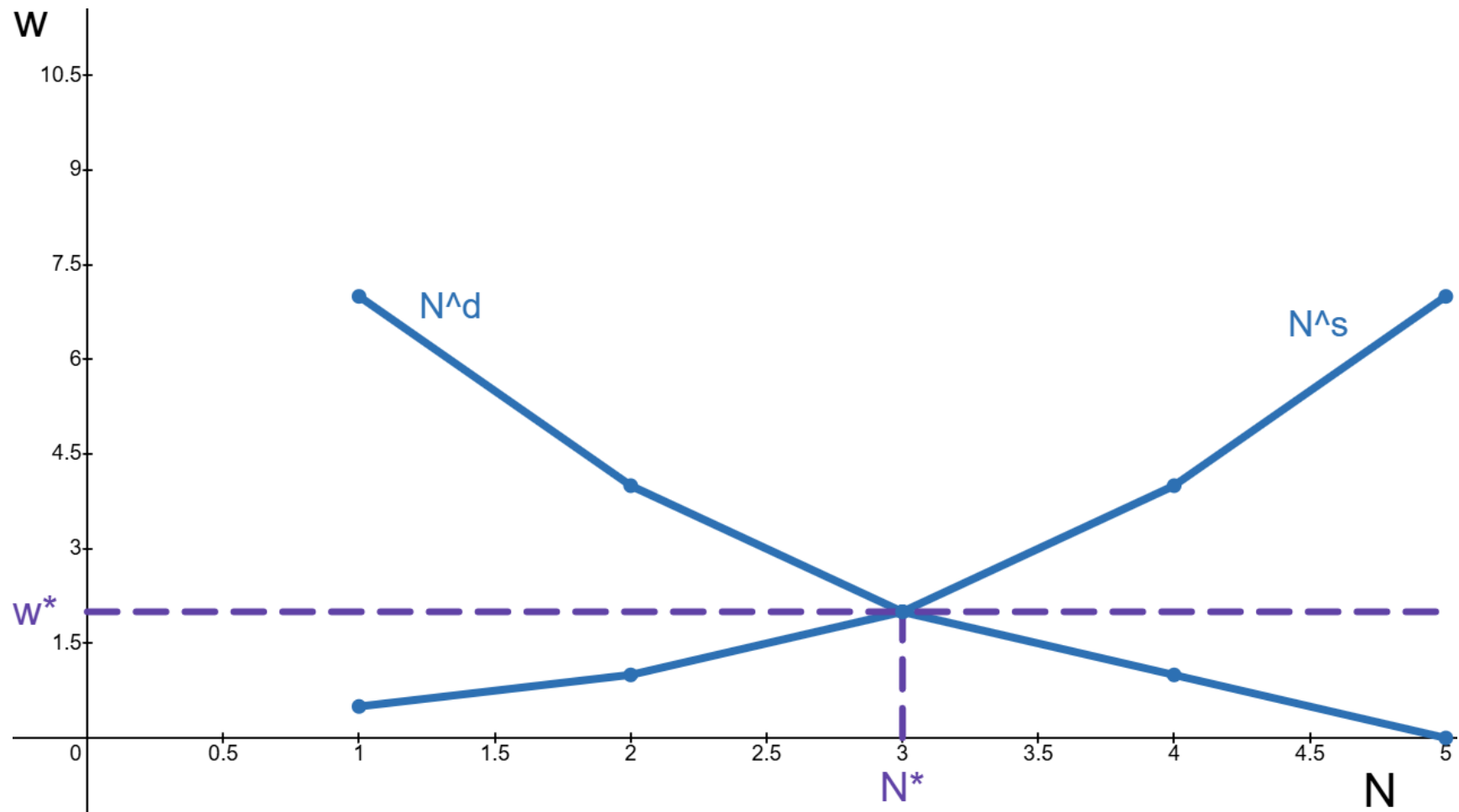
Market Clearing, again

- › **Market clearing:** price(s), demand function(s), and supply function(s) where quantity supplied = quantity demanded
 - Here, the quantity is labor and the price is the real wage
 - ❖ Real wage where quantity labor supplied = quantity labor demanded

- › Our example: **market clearing wage = 2**
 - Labor supplied = labor demanded

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Graphing Market Clearing



Changes in the Labor Market

- › Markets are **dynamic**; things can change
- › We can have **shocks** to supply, demand, and price
 - These shocks can also be dynamic, or permanent

Demand Shocks

- › A **demand shock** is anything that changes demand (for labor) at a given price (of labor)
 - Anything that changes the *MPN*

- › **Positive demand shocks (shift demand right)**
 1. Better technology (z)
 2. More capital (k)
 - What does this assume about capital and labor?

Example: Increase in k

- › An increase in k increases labor demand

Supply Shocks

- › A **supply shock** is anything that changes supply (of labor) at a given price (of labor)

- › **Positive supply shocks (shift supply right)**
 1. Increase in labor force
 - Raise social security age, more immigration, etc
 2. Lower wealth
 - Can't consume as much so $MU(c)$ increases
 3. Lower *future* wages
 - Similar reasons to #2. Note on this later

Example: Fall in Future Wages

- › Fall in future wages increases labor supply

The Y-axis Does NOT Shift Curves

- › Remember: whatever variable is on the y-axis, in this chapter it's the real wage (w), will NEVER cause a shift
- › *What we've been looking at is the current real wage (w)*
- › A change in the future wage (w') is not the y-axis variable and will therefore cause a shift
 - Technical terms: w' is not the argument to the N^s function

The Labor Force (BLS, 2026)

- › **Labor force:** 16+ yrs old, noninstitutional civilians
 - Employed
 - Unemployed
- › **Labor force participation rate** is roughly 63%
 - $(\text{labor force} \div \text{available population}) * 100$
- › **Unemployed:** actively looking for work in the last 4 weeks
- › **Unemployment rate** is roughly 5%
 - $(\text{total unemployed} \div \text{total labor force}) * 100$

Types of Unemployment

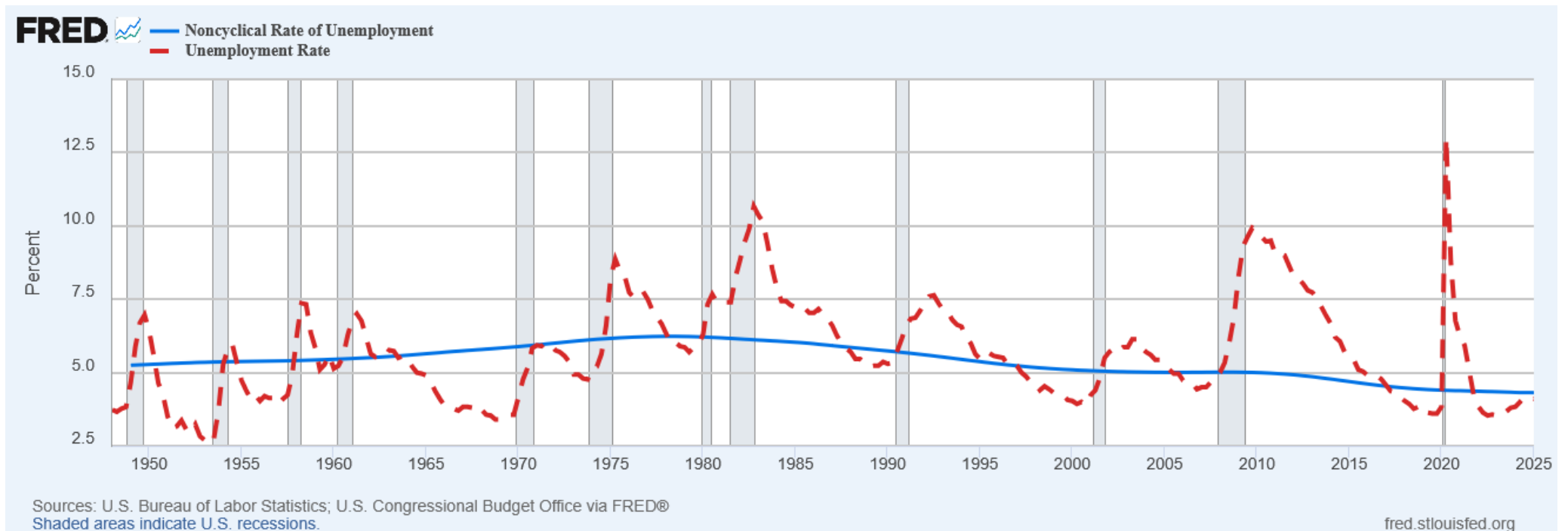
- › Frictional unemployment
 - Takes time to find a job
 - Usually, people looking for something that suits them
- › Structural unemployment
 - Skill mismatch: no one is hiring a VCR repair worker
 - Training them to be semiconductor manufactures takes time
- › Cyclical unemployment
 - Business cycles; unemployment is higher in recessions

Noncyclical Rate of Unemployment

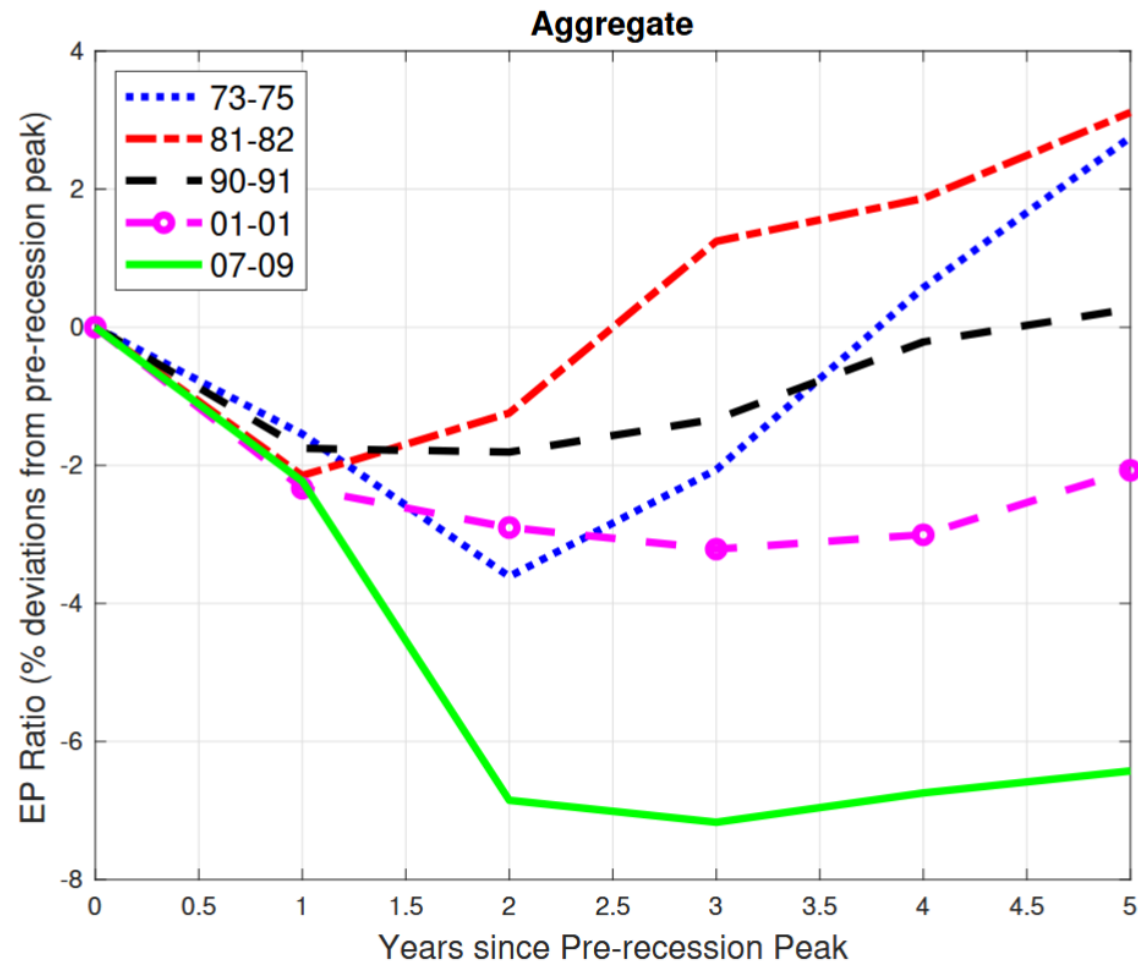
› Noncyclical Rate of Unemployment:

- Unemployment when cyclical rate = 0
- Can still have frictional & structural unemployment
- Used to be called the natural rate of unemployment
- Economy is said to be at **Full Employment**

Data: Noncyclical Rate of Unemployment

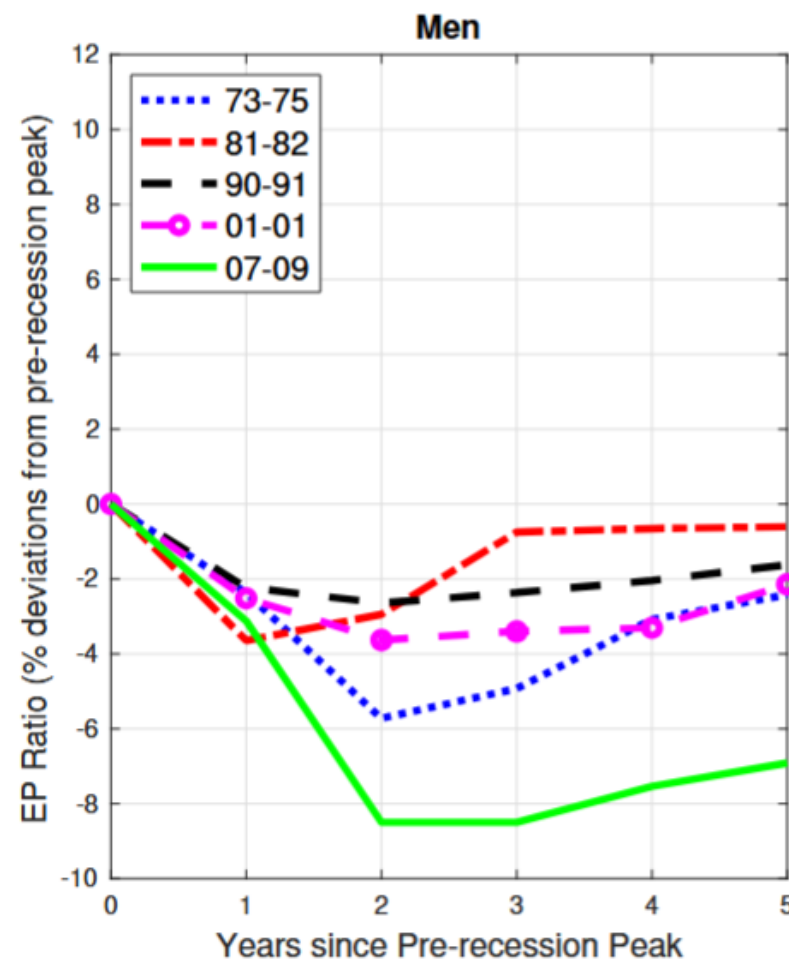
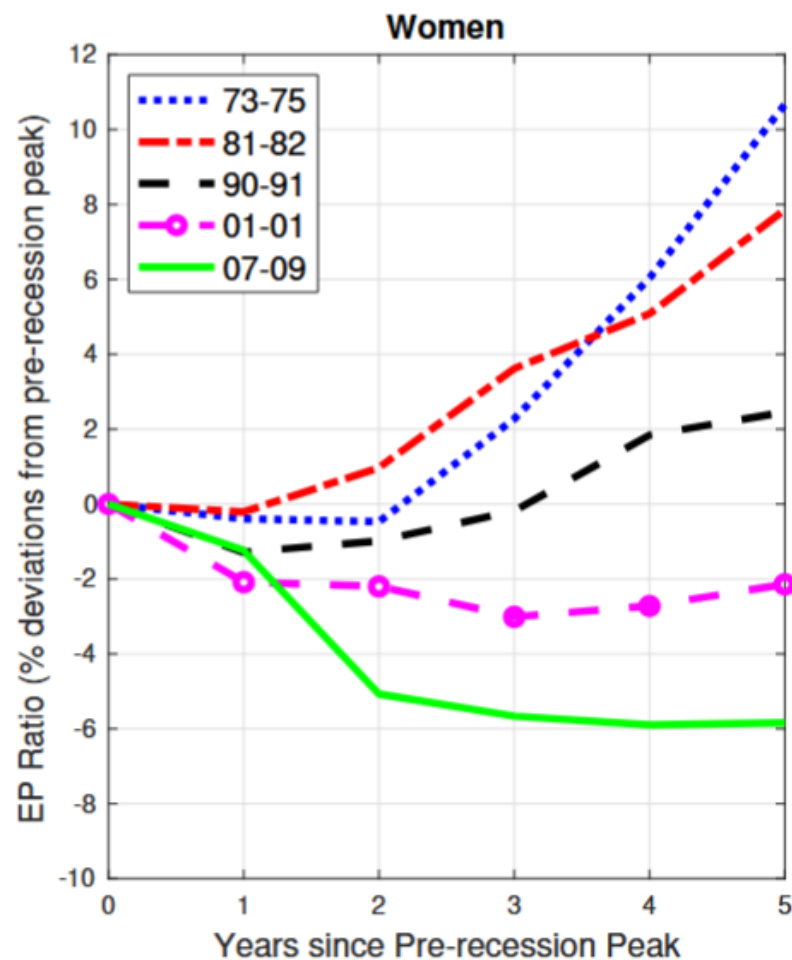


Jobless Recoveries?



(Chakraborty, 2022. Figure B3a)

Chakraborty (2022)



(Chakraborty, 2022. Figure B3b)

References

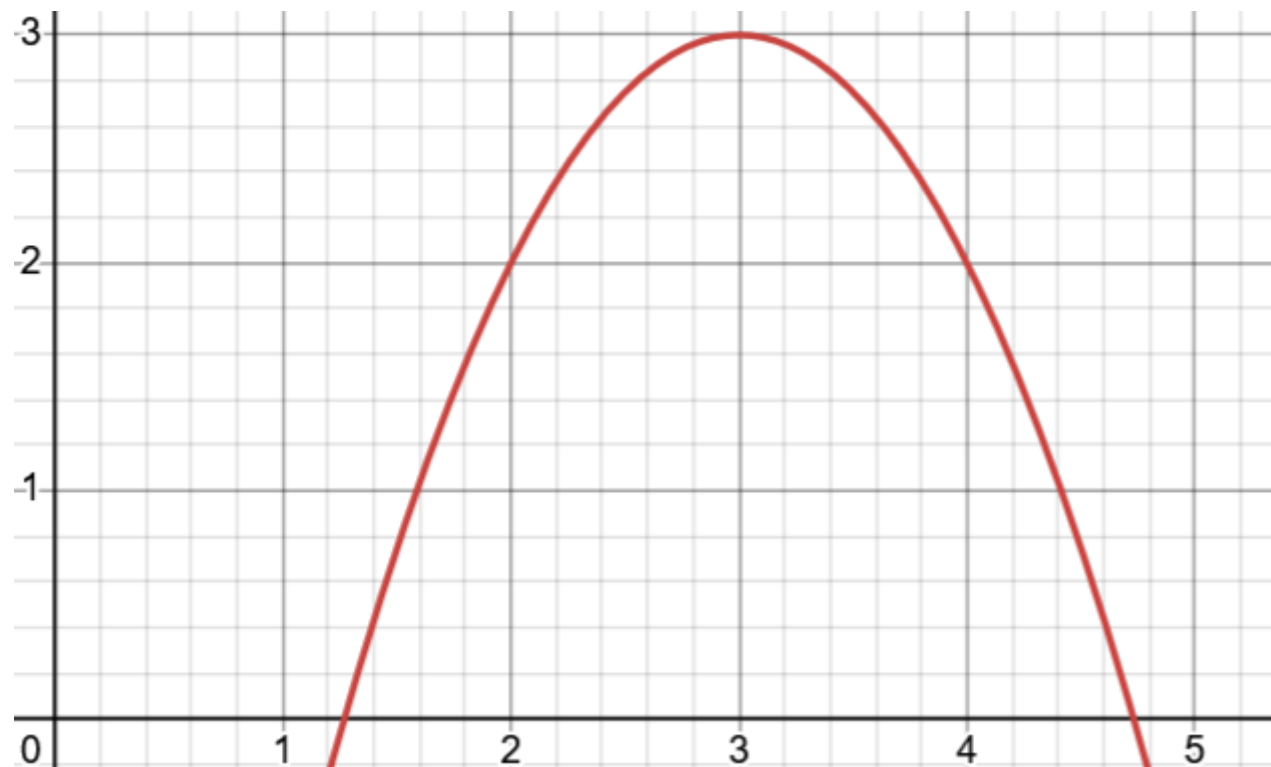
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- A. All non-cited graphs for this course made by me on [Desmos](#).

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APPENDIX

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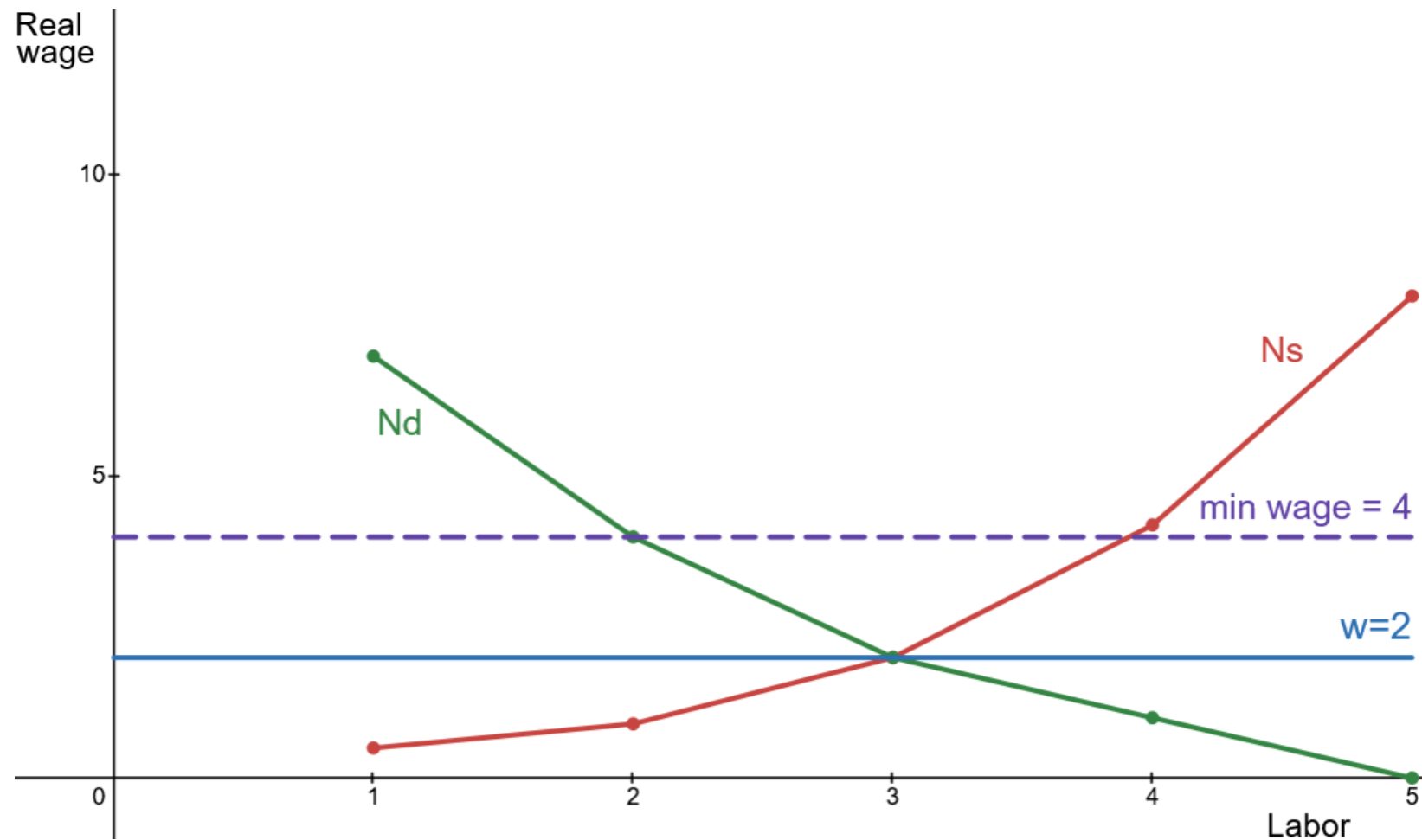
Why Two Equal Profit Values on Slide 22?



Minimum Wage

- › **Minimum wage** sets the lowest *legal* wage in the market
 - A change in the minimum wage can be a price shock
- › **Binding minimum wage** forces a change in the market
 - Leads to excess supply of labor; more people want to work than the market demands
 - In our example, any minimum (real) wage over 2 is binding
 - A minimum wage of 4 leads to an excess supply of roughly 2

Binding Minimum Wage



Minimum Wages and Employment (Card & Krueger, 1994)

- › NJ min wage went from \$4.25 to \$5.05 while PA had no change
- › Compare fast food employment on the border of the two states
 - Why on the border?
 - No significant changes (Card & Krueger, 1994)
- › So, a higher minimum wage does not lead to unemployment?
- › Well, yes and no
 - Many micro-level firm dynamics at play (two-year warning period)
 - Cannot say that larger increases will not cause unemployment
 - Prices of fast food went up to cover costs
 - Conflicts with other studies, no long-term analysis (employment or new firms)

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